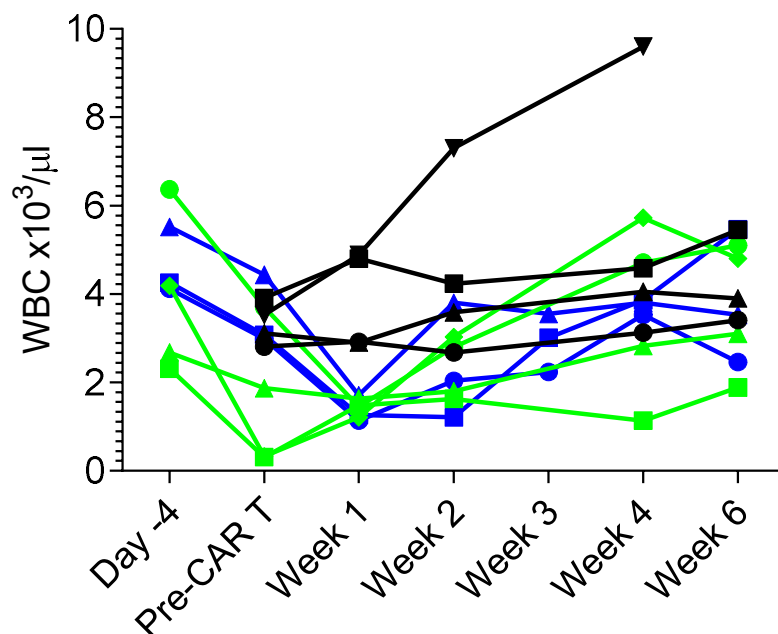


Supplemental Information

CAR T Cells Administered in Combination with Lymphodepletion and PD-1 Inhibition to Patients with Neuroblastoma

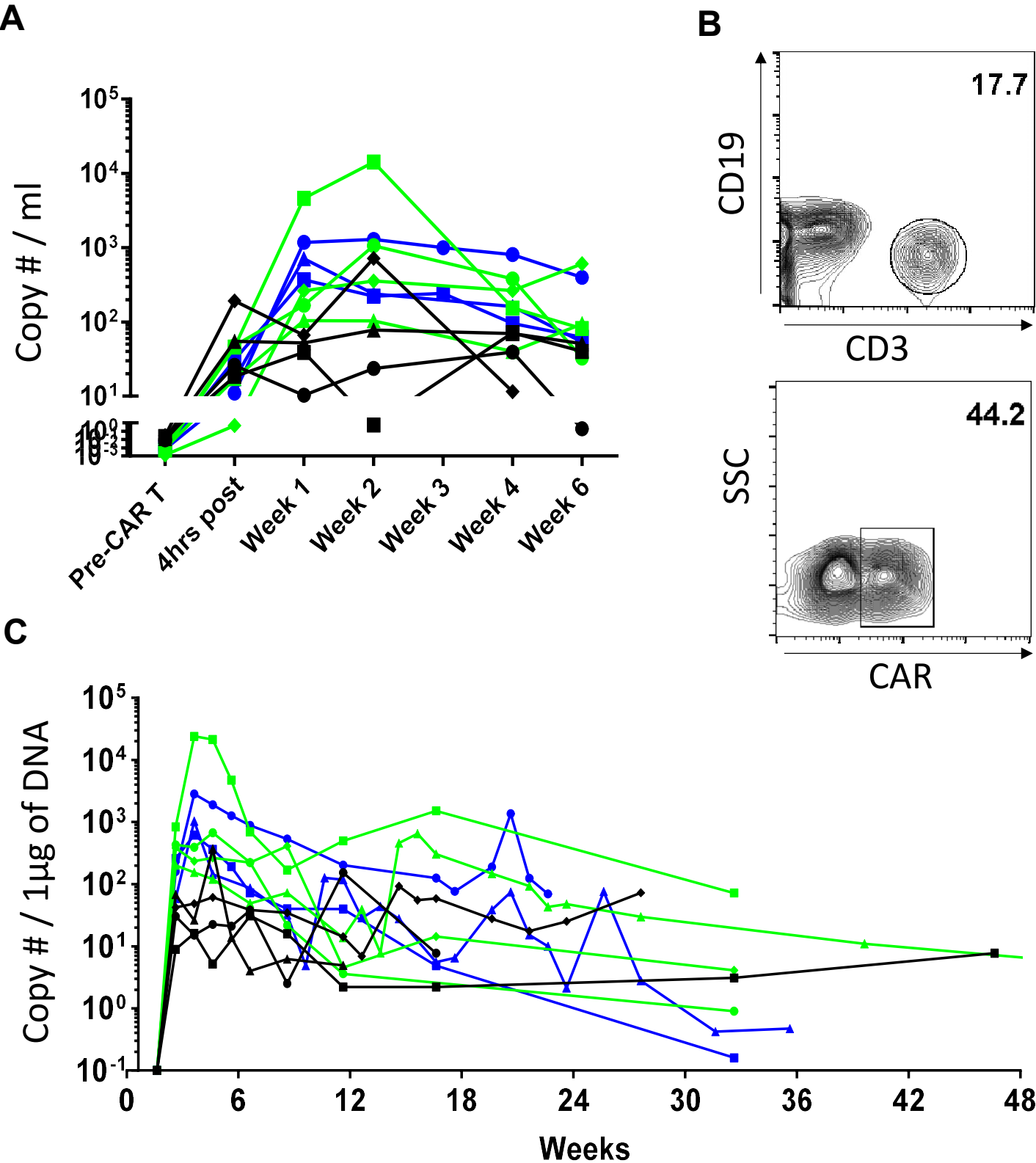
Andras Heczey, Chrystal U. Louis, Barbara Savoldo, Olga Dakhova, April Durett, Bambi Grilley, Hao Liu, Mengfeng F. Wu, Zhuyong Mei, Adrian Gee, Birju Mehta, Huimin Zhang, Nadia Mahmood, Haruko Tashiro, Helen E. Heslop, Gianpietro Dotti, Cliona M. Rooney, and Malcolm K. Brenner

Figure S1



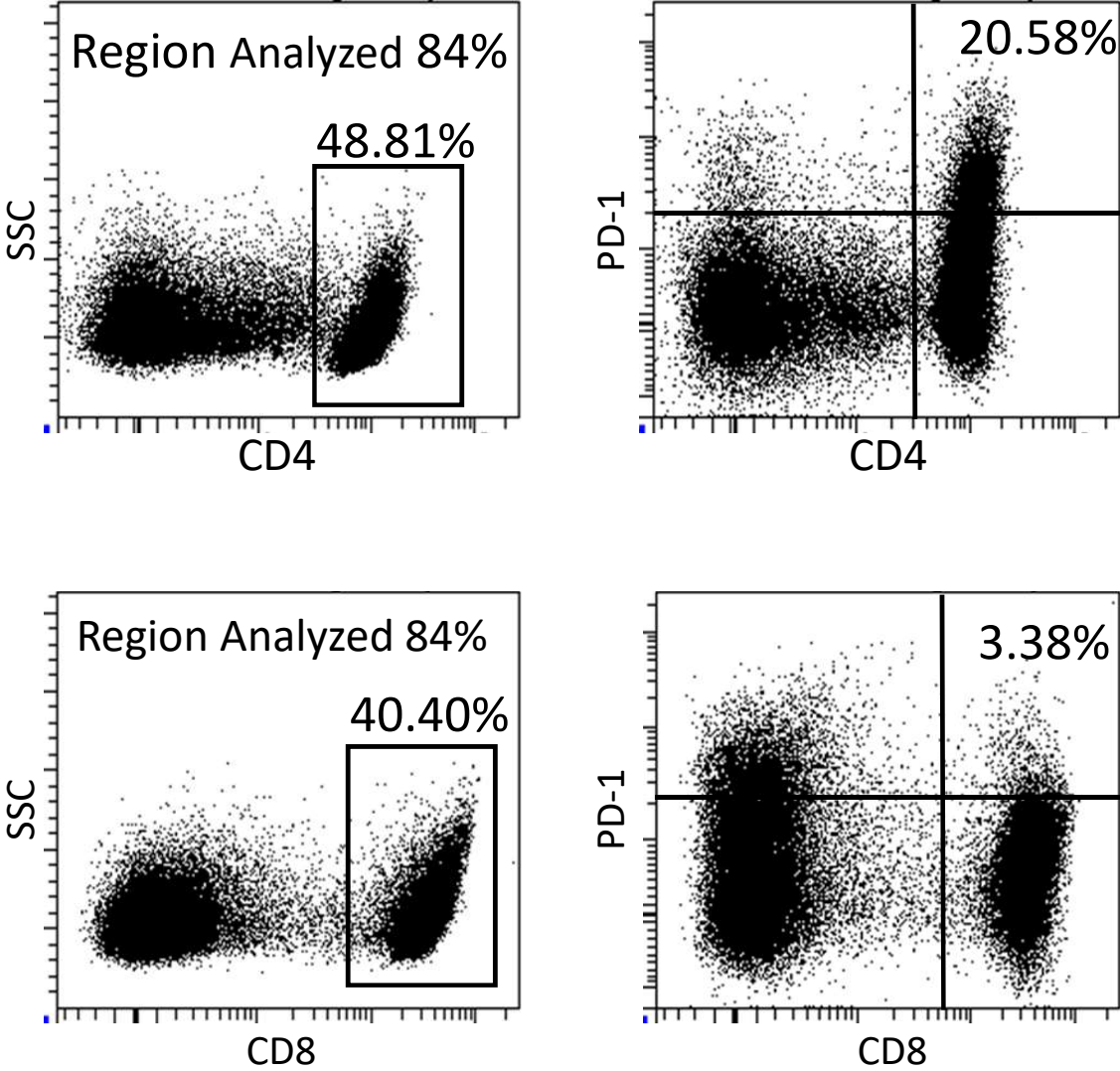
Supplementary Figure S1: Peripheral blood white blood cell count in patients receiving GD2-CAR3 T cells: White blood cell count $\times 10^3/\mu\text{l}$ measured at indicated time points and shown for each patient grouped by cohorts. Patients in cohort 1 are represented in black, cohort 2 in green and cohort 3 in blue; each patient per cohort is represented by a different shape.

Figure S2



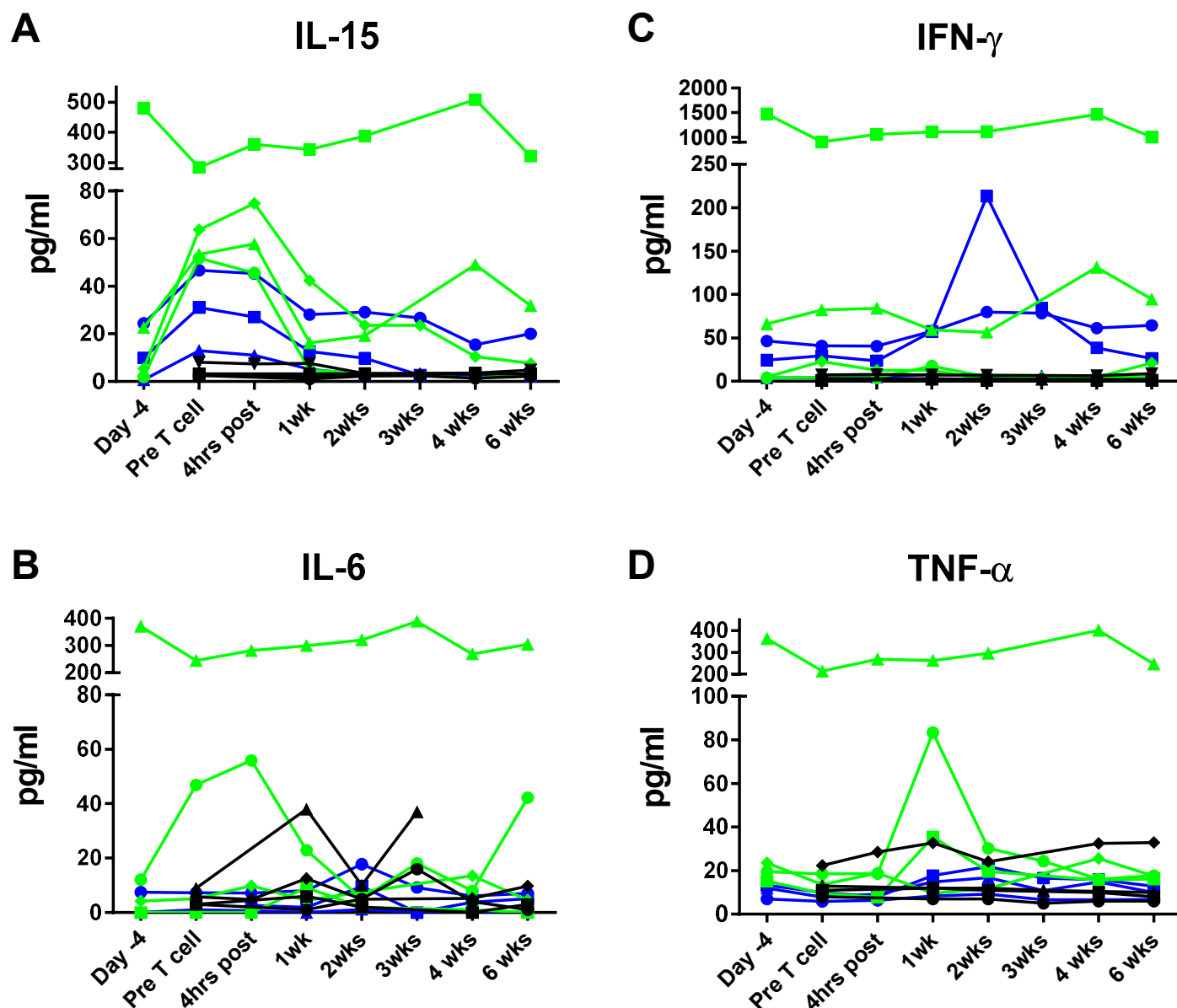
Supplementary Figure S2: Expansion and persistence of GD2-CAR3 T cells after adoptive transfer: Peripheral blood evaluation of patients infused with GD2-CAR3 T cells at indicated time points measured by RT-PCR. **A:** Expansion of GD2-CAR3 T cells in each patient during the initial 6 week monitoring period. **B:** CAR expression on CD3+ T cells in peripheral blood for patient 2A shown by FACS analysis. **C:** Long term persistence of GD2-CAR3 T cells in each patient measured by RT-PCR. For **A and C:** Patients in cohort 1 are represented in black, cohort 2 in green and cohort 3 in blue; each patient per cohort is represented by a different shape.

Figure S3

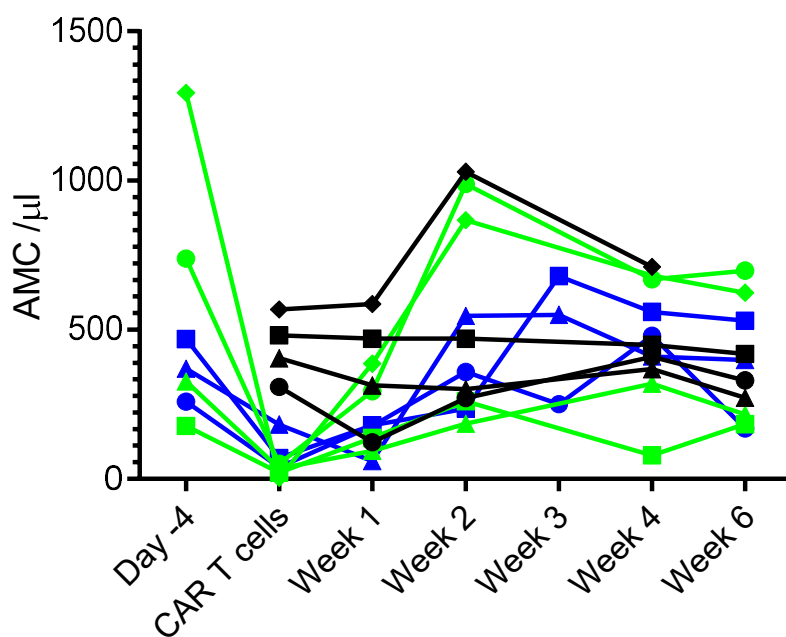


Supplementary Figure S3: PD-1 expression on CAR T cells of Patient 3C: PD-1 expression on CD4 and CD8 subsets of GD2-CAR3 T cells was evaluated by flow cytometry.

Figure S4

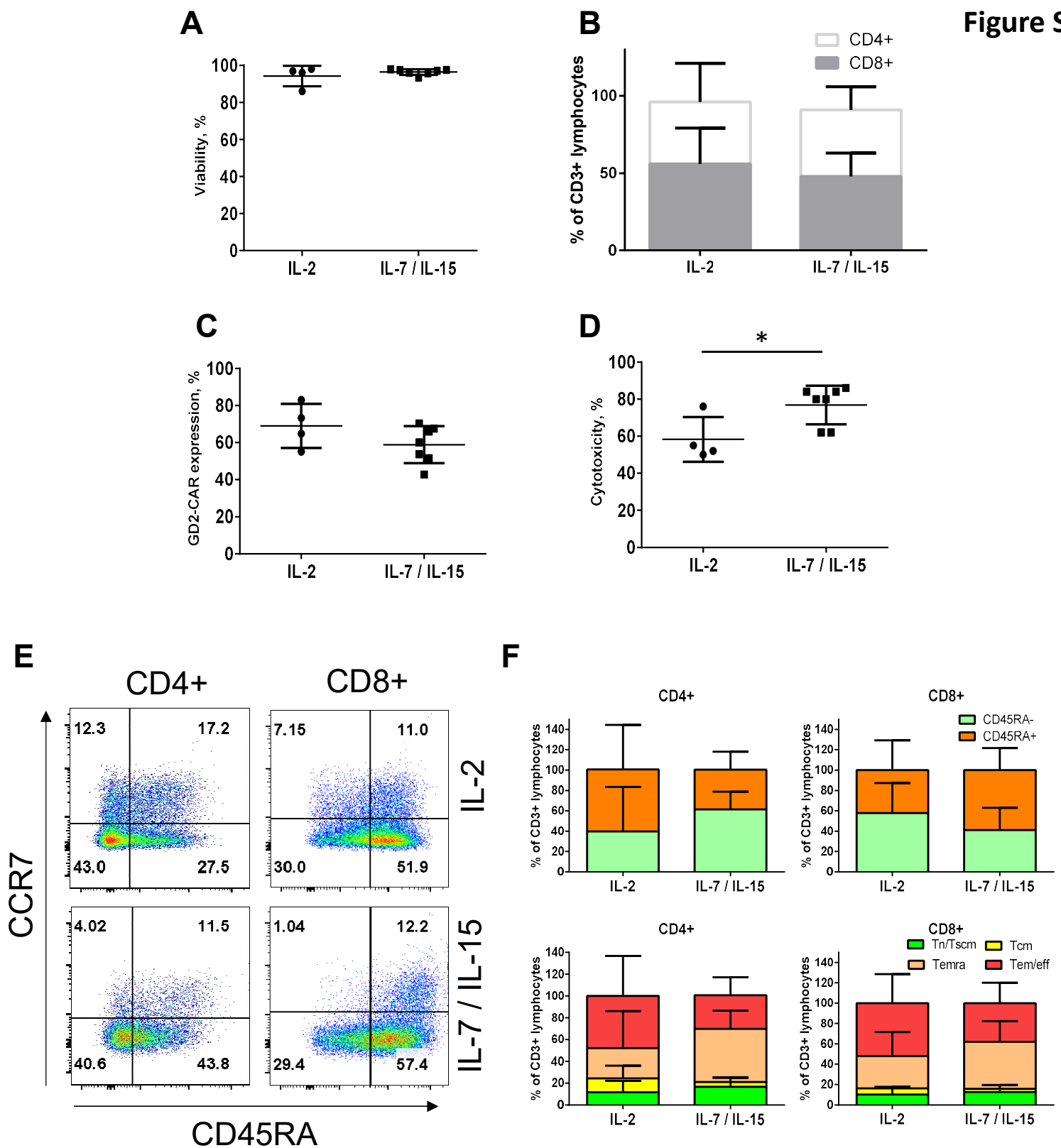


Supplementary Figure S4: Serum cytokines levels in patients receiving GD2-CAR3 T cells: Serum cytokine level was measured by Luminex™ assay at indicated time points and is shown for each patient grouped by cohorts. **A:** Interleukin-15 (IL-15). **B:** Interferon-γ (IFN-γ). **C:** Interleukin-6 (IL-6). **D:** Tumor necrosis factor – α (TNF-α). Patients in cohort 1 are represented in black, cohort 2 in green and cohort 3 in blue; each patient per cohort is represented by a different shape.



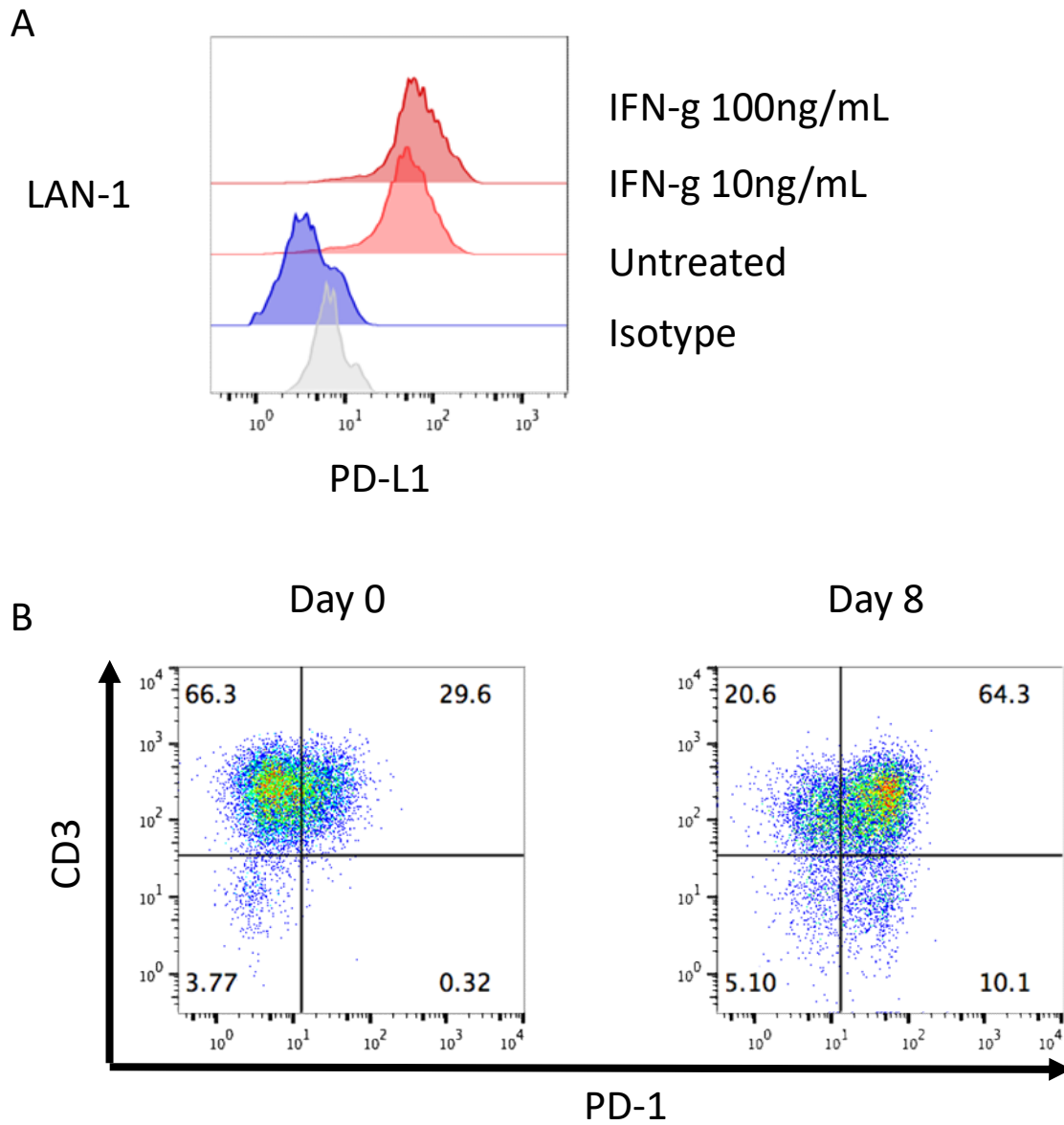
Supplementary Figure S5: Peripheral blood absolute monocyte count in patients receiving GD2-CAR3 T cells: Absolute monocyte count (AMC) $\times 10^3$ / μl measured at indicated time points and shown for each patient grouped by cohorts. Patients in cohort 1 are represented in black, cohort 2 in green and cohort 3 in blue; each patient per cohort is represented by a different shape.

Figure S6



Supplementary Figure S6: Effect of homeostatic cytokines during manufacturing on GD2-CAR3 T cell phenotype: Comparison of GD2-CAR3 T cell products manufactured with IL-2 or IL-7/15 by flow cytometry. **A:** The viability of GD2-CAR3 T cells products measured by 7AAD staining at the end of manufacturing. **B:** Percentages of CD4 and CD8 subsets of CD3+ T cells. **C:** Percentage of GD2-CAR3+ T cells in infusion products. **D:** Cytotoxicity of GD2-CAR3 T cell products against CHLA255 NB cell line by standard 4hr ^{51}Cr release assay. *: $p < 0.05$, **: $p < 0.01$ and ***: $p < 0.001$. **E:** Representative flow cytometry dot plots of CD4 and CD8 gated CD45RA/CCR7 T cells subsets. **F:** Percentages of naive/stem cell memory (Tn/scm, CD45RA+/CCR7+), central memory (CM, CD45RA-/CCR7+), effector memory/effector (Em/Eff, CD45RA-/CCR7-) and effector memory CD45RA positive (Emra, CD45RA+/CCR7-) T cells in CD4 and CD8 T cell subsets.

Figure S7



Supplementary Figure S7: Expression of PD-L1 on neuroblasts and PD-1 on CAR T cells:

A: To test the effect of interferon- γ (IFN- γ) on neuroblasts, LAN-1 neuroblastoma cell line was incubated with indicated concentrations of IFN- γ and PD-L1 expression was evaluated at 48hrs by flow cytometry. **B:** To test the effect of activation by CAR, GD2-CAR3 T cells were co-cultured with LAN-1 neuroblasts and PD1 expression was evaluated before and on Day 8 by flow cytometry.